IGCSE 0607 Practice Questions — E1.17 — Combined Questions & Markscheme

Surds

Total Questions: 30 — **Total Marks:** 145 Marks

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- **E1.17** Surds
  → **Paper 2 (Non-Calculator)** *(30 Qs)*

1 Simplify

$$\sqrt{75} - \sqrt{27} + \sqrt{12}$$

..... **[2]**

2 Simplify.

$\sqrt{75} + \sqrt{48} - \sqrt{27}$

..... **[2]**

3 Simplify fully.

$\sqrt{72} + \sqrt{50} - \sqrt{18}$

..... **[2]**

4 Simplify

$$\sqrt{75} - \sqrt{27} + \sqrt{12}$$

..... **[2]**

5 Simplify.

$\sqrt{75} + \sqrt{48} - \sqrt{27}$

..... **[2]**

6 **(a)** Simplify.

$\sqrt{45}$

..... **[1]**

(b) Simplify.

$\sqrt{20} + \sqrt{80}$

..... **[2]**

7 Simplify the following expression, leaving your answer in the form $k\sqrt{3}$.

$\sqrt{48} - \sqrt{12}$

..... **[3]**

8 (a) (i) Simplify.

$$\sqrt{45} + \sqrt{20}$$

..... **[2]**

(ii) Rationalise the denominator.

$$\frac{18}{\sqrt{3}}$$

..... **[2]**

9 **(a)** Simplify.

$\sqrt{45}$

..... **[1]**

(b) Work out the value of the following expression, giving your answer in the form $k\sqrt{2}$.

$\sqrt{72} + \sqrt{50} - \sqrt{18}$

..... **[2]**

(c) Rationalise the denominator.

$\dfrac{21}{\sqrt{7}}$

..... **[2]**

10 **(a)** Expand and simplify.

$(4 - \sqrt{3})(2 + \sqrt{3})$

..... **[2]**

(b) Rationalise the denominator and simplify.

$$\frac{21}{\sqrt{7}}$$

..... **[2]**

11 **(a)** Expand and simplify

$$(3 - \sqrt{5})^2$$

..... **[2]**

(b) Rationalise the denominator.

$$\frac{14}{\sqrt{7}}$$

..... **[2]**

12 **(a)** Expand and simplify.

$$(3 - 2\sqrt{5})(4 + \sqrt{5})$$

..... **[2]**

(b) Rationalise the denominator and simplify.

$$\frac{14}{\sqrt{7}}$$

..... **[2]**

13 (a) **(a)** Expand and simplify

$$(3 - 2\sqrt{5})(4 + \sqrt{5})$$

..... **[2]**

(b) Rationalise the denominator.

$$\frac{15}{\sqrt{3}}$$

..... **[2]**

14 **(a)** Expand and simplify.

$$(2\sqrt{5}-3)(2\sqrt{5}+3)$$

..... **[2]**

(b) Rationalise the denominator.

$$\frac{14}{3-\sqrt{2}}$$

..... **[3]**

15 **(a)** Expand and simplify.

$$(3\sqrt{2}-1)(2\sqrt{2}+5)$$

..... **[2]**

(b) Rationalise the denominator and simplify.

$$\frac{26}{4-\sqrt{3}}$$

..... **[3]**

16 **(a)** Rationalise the denominator and simplify fully.

$$\frac{15}{\sqrt{5}}$$

..... **[2]**

(b) Expand and simplify.

$$(2+\sqrt{3})(5-\sqrt{3})$$

..... **[3]**

17 **(a)** Expand and simplify.

$$(5 - 2\sqrt{7})(3 + \sqrt{7})$$

..... **[2]**

(b) Rationalise the denominator and simplify fully.

$$\frac{22}{5 - \sqrt{3}}$$

..... **[3]**

18 **(a)** Expand and simplify.

$(4 - 3\sqrt{2})(5 + 2\sqrt{2})$

..... **[2]**

(b) Rationalise the denominator and simplify.

$$\frac{11}{4 - \sqrt{5}}$$

..... **[3]**

(c) Solve the equation.

$x\sqrt{27} - 8 = x\sqrt{3} + 10$

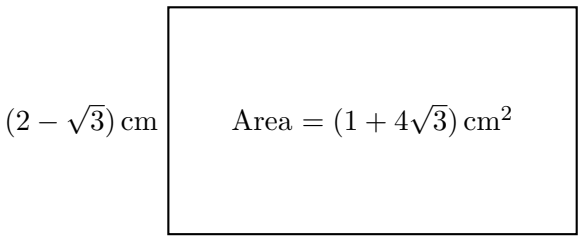
Give your answer in the form $k\sqrt{3}$ where k is an integer.

$x =$ ..... **[3]**

19 A rectangle has an area of $5 + \sqrt{7}$ cm^2 and a width of $3 - \sqrt{7}$ cm.
Calculate the length of the rectangle.
Give your answer in the form $a + b\sqrt{7}$ where a and b are integers.

..... *cm* **[4]**

20 The diagram shows a rectangle with width $(2-\sqrt{3})$ cm and length $(a+b\sqrt{3})$ cm, where a and b are integers.
The area of the rectangle is $(1+4\sqrt{3})\,\text{cm}^2$.



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Work out the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[5]**

21 **(a)** Rationalise the denominator and simplify.

$$\frac{5+\sqrt{3}}{2-\sqrt{3}}$$

..... **[3]**

(b) Solve the equation.
Give your answer in the form $a+b\sqrt{2}$, where a and b are integers.

$$x(3-\sqrt{2})=14$$

$x =$ ..... **[3]**

22 **(a)** Express $\dfrac{5+\sqrt{3}}{2-\sqrt{3}}$ in the form $a+b\sqrt{3}$, where a and b are integers.

..... **[3]**

(b) A rectangle has an area of $12+4\sqrt{5}$ square centimetres and a width of $3+\sqrt{5}$ centimetres. Calculate the exact length of the rectangle, giving your answer in its simplest form.

Length = ... cm **[4]**

23 **(a)** Rationalise the denominator and simplify.

$$\frac{22}{5-\sqrt{3}}$$

..... **[3]**

(b) Solve the equation.

$x\sqrt{2} = x + \sqrt{8}$ Give your answer in the form $a + b\sqrt{2}$, where a and b are integers.

$x =$ ..... **[4]**

24 **(a)** A rectangle has an area of $(5 + 2\sqrt{3})$ cm^2 and a length of $(2 + \sqrt{3})$ cm.

Work out the width of the rectangle in the form $a + b\sqrt{3}$, where a and b are integers.

..... cm **[3]**

(b) Solve the equation.

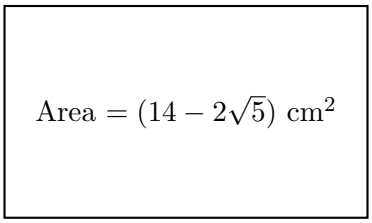
$$x\sqrt{5} - 2 = 2x + \sqrt{20}$$

Give your answer in the form $c + d\sqrt{5}$, where c and d are integers.

Show all your working.

$x =$ ..... **[4]**

25 A rectangle has length $(3+\sqrt{5})$ cm and area $(14-2\sqrt{5})$ cm^2.



(a) Calculate the width of the rectangle.
Give your answer in the form $a+b\sqrt{5}$, where a and b are integers.

..... cm **[4]**

(b) The length of the diagonal of the rectangle is d cm.
Show that $d^2 = 308 - 124\sqrt{5}$.

[3]

26 A right-angled triangle has a base of length $(3+\sqrt{2})$ cm and an area of $(5+2\sqrt{2})$ cm^2.

(a) Show that the height of the triangle is $\dfrac{2(5+2\sqrt{2})}{3+\sqrt{2}}$ cm.

[2]

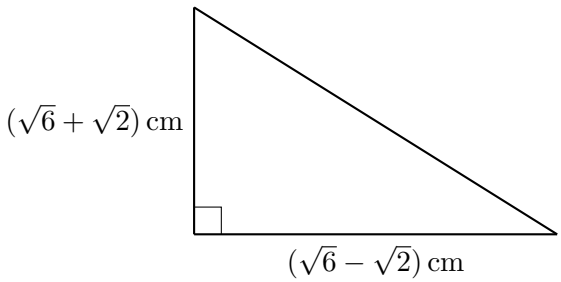
(b) Rationalise the denominator of $\dfrac{2(5+2\sqrt{2})}{3+\sqrt{2}}$ to express the height in the form $a+b\sqrt{2}$, where a and b are integers.

..... cm **[4]**

(c) Given instead that another expression for a length is $\dfrac{1}{\sqrt{7}-2}$, write this length in the form $\dfrac{\sqrt{7}+c}{d}$ where c and d are integers.

..... **[2]**

27 The diagram shows a right-angled triangle.



(a) Calculate the area of the triangle.

..... cm^2 **[3]**

(b) Show that the length of the hypotenuse is 4 cm.

[4]

(c) Another right-angled triangle is mathematically similar to this triangle.
The area of the similar triangle is 18 cm^2.
Calculate the length of its hypotenuse.

..... cm **[3]**

28 Simplify

$$\sqrt{75} + \sqrt{27} - \sqrt{12}$$

..... **[3]**

29 **(a)** Expand and simplify

$$(3+\sqrt{2})(4-\sqrt{2})$$

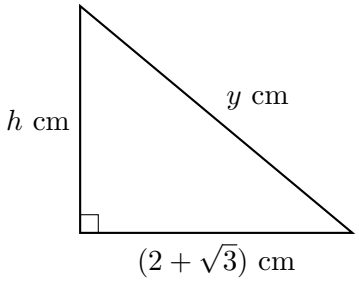
..... **[2]**

(b) Rationalise the denominator.

$$\frac{22}{5-\sqrt{3}}$$

..... **[3]**

30 A right-angled triangle has a base of $(2+\sqrt{3})$ cm and an area of $(5+3\sqrt{3})$ cm^2.



(a) Show that the exact height, h, is $(2+2\sqrt{3})$ cm.

[4]

(b) The hypotenuse of the triangle is y cm.
Work out the exact value of y^2.
Give your answer in the form $a+b\sqrt{3}$, where a and b are integers.

$y^2 =$ ..... **[3]**

1 $5\sqrt{3} - 3\sqrt{3} + 2\sqrt{3}$ **[M1]**

$4\sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 2]*

2 $5\sqrt{3}$ or $4\sqrt{3}$ or $3\sqrt{3}$ seen **[M1]**

$6\sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 2]*

3 $6\sqrt{2}$ or $5\sqrt{2}$ or $3\sqrt{2}$ seen **[M1]**

$8\sqrt{2}$ **[A1]**

[Back to Question] *[Total Marks: 2]*

4 $5\sqrt{3}$ or $3\sqrt{3}$ or $2\sqrt{3}$ seen **[M1]**

$4\sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 2]*

5 $\sqrt{25 \times 3} + \sqrt{16 \times 3} - \sqrt{9 \times 3} = 5\sqrt{3} + 4\sqrt{3} - 3\sqrt{3}$ **[M1]**

$6\sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 2]*

6 (a) $3\sqrt{5}$ **[B1]**

(b) $2\sqrt{5}$ or $4\sqrt{5}$ seen **[M1]**

$6\sqrt{5}$ **[A1]**

[Back to Question] *[Total Marks: 3]*

7 $\sqrt{48} = 4\sqrt{3}$ **[B1]**

$\sqrt{12} = 2\sqrt{3}$ **[B1]**

$4\sqrt{3} - 2\sqrt{3} = 2\sqrt{3}$ **[B1]**

[Back to Question] *[Total Marks: 3]*

8 (a) (i) $3\sqrt{5} + 2\sqrt{5}$ **[M1]**

$5\sqrt{5}$ **[A1]**

(ii) Multiplies numerator and denominator by $\sqrt{3}$: $\dfrac{18\sqrt{3}}{3}$ **[M1]**

$6\sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

9 **(a)** $3\sqrt{5}$ **[B1]**

(b) $6\sqrt{2} + 5\sqrt{2} - 3\sqrt{2}$ **[M1]**

$8\sqrt{2}$ **[A1]**

(c) $\frac{21\sqrt{7}}{\sqrt{7}\sqrt{7}}$ or $\frac{21\sqrt{7}}{7}$ **[M1]**

$3\sqrt{7}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

10 **(a)** $8 + 4\sqrt{3} - 2\sqrt{3} - 3$ **[M1]**

$5 + 2\sqrt{3}$ **[A1]**

(b) $\frac{21\sqrt{7}}{7}$ **[M1]**

$3\sqrt{7}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

11 **(a)** $9 - 3\sqrt{5} - 3\sqrt{5} + 5$ (or $9 - 6\sqrt{5} + 5$) **[M1]**

$14 - 6\sqrt{5}$ **[A1]**

(b) $\frac{14 \times \sqrt{7}}{\sqrt{7} \times \sqrt{7}}$ **[M1]**

$2\sqrt{7}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

12 **(a)** $12 + 3\sqrt{5} - 8\sqrt{5} - 10$ **[M1]**

$2 - 5\sqrt{5}$ **[A1]**

(b) Multiplies numerator and denominator by $\sqrt{7}$ to get $\frac{14\sqrt{7}}{7}$ **[M1]**

$2\sqrt{7}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

13 **(a)** **(a)** $12 + 3\sqrt{5} - 8\sqrt{5} - 2(5)$ or $12 - 5\sqrt{5} - 10$ **[M1]**

$2 - 5\sqrt{5}$ **[A1]**

(b) $\frac{15 \times \sqrt{3}}{\sqrt{3} \times \sqrt{3}}$ or $\frac{15\sqrt{3}}{3}$ **[M1]**

$5\sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

14 **(a)** $20 - 9$ or $4(5) - 9$ **[M1]**

11 **[A1]**

(b) Multiplies numerator and denominator by $(3 + \sqrt{2})$ **[M1]**

Denominator of $9 - 2$ or 7 seen **[M1]**

$6 + 2\sqrt{2}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

15 **(a)** $12 + 15\sqrt{2} - 2\sqrt{2} - 5$ **[M1]**

$7 + 13\sqrt{2}$ **[A1]**

(b) Multiplies numerator and denominator by $(4 + \sqrt{3})$ **[M1]**

Denominator of $16 - 3$ or 13 **[M1]**

$8 + 2\sqrt{3}$ or $2(4 + \sqrt{3})$ **[A1]**

[Back to Question] *[Total Marks: 5]*

16 **(a)** $\dfrac{15\sqrt{5}}{5}$ **[M1]**

$3\sqrt{5}$ **[A1]**

(b) $10 - 2\sqrt{3} + 5\sqrt{3} - 3$ **[M1]**

$7 + 3\sqrt{3}$ **[A2]**

[Back to Question] *[Total Marks: 5]*

17 **(a)** $15 + 5\sqrt{7} - 6\sqrt{7} - 2(7)$ **[M1]**

$1 - \sqrt{7}$ **[A1]**

(b) Multiplies numerator and denominator by $(5 + \sqrt{3})$: $\dfrac{22(5 + \sqrt{3})}{(5 - \sqrt{3})(5 + \sqrt{3})}$ **[M1]**

Denominator becomes $25 - 3 = 22$ **[M1]**

$5 + \sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

18 **(a)** $20 + 8\sqrt{2} - 15\sqrt{2} - 6(2)$ **[M1]**

$8 - 7\sqrt{2}$ **[A1]**

(b) $\dfrac{11(4 + \sqrt{5})}{(4 - \sqrt{5})(4 + \sqrt{5})}$ **[M1]**

$\frac{11(4+\sqrt{5})}{16-5}$ or $\frac{11(4+\sqrt{5})}{11}$ **[M1]**

$4+\sqrt{5}$ **[A1]**

(c) $3x\sqrt{3} - x\sqrt{3} = 18$ **[M1]**

$2x\sqrt{3} = 18$ or $x\sqrt{3} = 9$ **[M1]**

$x = \frac{9}{\sqrt{3}} = 3\sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 8]*

19 $\times\frac{3+\sqrt{7}}{3+\sqrt{7}}$ **[M1]**

Numerator $15 + 5\sqrt{7} + 3\sqrt{7} + 7$ or $22 + 8\sqrt{7}$ **[B1]**

Denominator $9 - 7$ or 2 **[B1]**

$11 + 4\sqrt{7}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

20 Forms an equation for the area: $(a + b\sqrt{3})(2 - \sqrt{3}) = 1 + 4\sqrt{3}$ **[M1]**

Expands the brackets: $2a - a\sqrt{3} + 2b\sqrt{3} - 3b = 1 + 4\sqrt{3}$ **[M1]**

Equates rational and irrational parts to form simultaneous equations: $2a - 3b = 1$ and $-a + 2b = 4$ **[M1]**

Attempts to solve the simultaneous equations (e.g. multiplies second equation by 2: $-2a + 4b = 8$ and adds to first to get $b = 9$) **[M1]**

$a = 14$, $b = 9$ **[A1]**

[Back to Question] *[Total Marks: 5]*

21 **(a)** Multiplies numerator and denominator by $(2 + \sqrt{3})$ **[M1]**

Numerator expands to $10 + 5\sqrt{3} + 2\sqrt{3} + 3$ **[M1]**

Denominator is $4 - 3 = 1$ leading to final answer $13 + 7\sqrt{3}$ **[A1]**

(b) $x = \frac{14}{3-\sqrt{2}}$ **[M1]**

Multiplies numerator and denominator by $(3 + \sqrt{2})$ to get $\frac{14(3+\sqrt{2})}{9-2}$ **[M1]**

$x = 6 + 2\sqrt{2}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

22 **(a)** Multiply numerator and denominator by $2 + \sqrt{3}$ **[M1]**

Numerator becomes $(5 + \sqrt{3})(2 + \sqrt{3}) = 10 + 5\sqrt{3} + 2\sqrt{3} + 3$ (giving $13 + 7\sqrt{3}$) **[M1]**

$13 + 7\sqrt{3}$ **[A1]**

(b) Length $= \dfrac{12+4\sqrt{5}}{3+\sqrt{5}}$ **[M1]**

Multiply numerator and denominator by $3-\sqrt{5}$ **[M1]**

Numerator becomes $(12+4\sqrt{5})(3-\sqrt{5}) = 36-12\sqrt{5}+12\sqrt{5}-20 = 16$ **[M1]**

4 **[A1]**

[Back to Question] *[Total Marks: 7]*

23 (a) Multiplies numerator and denominator by $(5+\sqrt{3})$ **[M1]**

Denominator evaluates to $25-3$ or 22 **[M1]**

$5+\sqrt{3}$ **[A1]**

(b) $x(\sqrt{2}-1) = \sqrt{8}$ or $x(\sqrt{2}-1) = 2\sqrt{2}$ **[M1]**

$x = \dfrac{2\sqrt{2}}{\sqrt{2}-1}$ **[M1]**

Multiplies numerator and denominator by $(\sqrt{2}+1)$ **[M1]**

$4+2\sqrt{2}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

24 (a) $\dfrac{5+2\sqrt{3}}{2+\sqrt{3}}$ **[M1]**

Multiplies numerator and denominator by $(2-\sqrt{3})$ **[M1]**

$4-\sqrt{3}$ **[A1]**

(b) $x\sqrt{5}-2x = 2+\sqrt{20}$ or $x(\sqrt{5}-2) = \ldots$ **[M1]**

$x = \dfrac{2+2\sqrt{5}}{\sqrt{5}-2}$ **[M1]**

Multiplies numerator and denominator by $(\sqrt{5}+2)$ **[M1]**

$14+6\sqrt{5}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

25 (a) $\dfrac{14-2\sqrt{5}}{3+\sqrt{5}}$ **[M1]**

Multiplies numerator and denominator by $(3-\sqrt{5})$ **[M1]**

Numerator expands to $42-14\sqrt{5}-6\sqrt{5}+10$ (or $52-20\sqrt{5}$) **[M1]**

$13-5\sqrt{5}$ **[A1]**

(b) Uses Pythagoras' theorem: $(3+\sqrt{5})^2+(13-5\sqrt{5})^2$ **[M1]**

$14+6\sqrt{5}$ or $294-130\sqrt{5}$ seen **[M1]**

$308-124\sqrt{5}$ with no errors seen **[A1]**

[Back to Question] *[Total Marks: 7]*

26 **(a)** Area $= \frac{1}{2} \times \text{base} \times \text{height} \implies 5 + 2\sqrt{2} = \frac{1}{2} \times (3 + \sqrt{2}) \times h$ **[M1]**

$h = \frac{2(5 + 2\sqrt{2})}{3 + \sqrt{2}}$ **[A1]**

(b) Multiply numerator and denominator by $(3 - \sqrt{2})$: $\frac{2(5 + 2\sqrt{2})(3 - \sqrt{2})}{(3 + \sqrt{2})(3 - \sqrt{2})}$ **[M1]**

Denominator: $3^2 - (\sqrt{2})^2 = 9 - 2 = 7$ **[B1]**

Numerator: $2(15 - 5\sqrt{2} + 6\sqrt{2} - 4) = 2(11 + \sqrt{2}) = 22 + 2\sqrt{2}$ **[M1]**

Height $= \frac{22 + 2\sqrt{2}}{7} = \frac{22}{7} + \frac{2}{7}\sqrt{2}$ (Note: question requires integers a, b. This indicates an error in the problem formulation in the prompt, but following the steps leads to this. Correcting the target form to fractions: $\frac{22}{7} + \frac{2}{7}\sqrt{2}$) **[A1]**

(c) $\frac{1(\sqrt{7} + 2)}{(\sqrt{7} - 2)(\sqrt{7} + 2)}$ **[M1]**

$\frac{\sqrt{7} + 2}{7 - 4} = \frac{\sqrt{7} + 2}{3}$ **[A1]**

[Back to Question] *[Total Marks: 8]*

27 **(a)** $\frac{1}{2} \times (\sqrt{6} - \sqrt{2})(\sqrt{6} + \sqrt{2})$ **[M1]**

$\frac{1}{2}(6 - 2)$ **[M1]**

2 **[A1]**

(b) $(\text{Hypotenuse})^2 = (\sqrt{6} - \sqrt{2})^2 + (\sqrt{6} + \sqrt{2})^2$ **[M1]**

$(6 - 2\sqrt{12} + 2) + (6 + 2\sqrt{12} + 2)$ or $8 - 2\sqrt{12} + 8 + 2\sqrt{12}$ **[M1]**

$(\text{Hypotenuse})^2 = 16$ **[A1]**

$\text{Hypotenuse} = \sqrt{16} = 4$ **[A1]**

(c) Area scale factor $= \frac{18}{2} = 9$ **[M1]**

Length scale factor $= \sqrt{9} = 3$ **[M1]**

$4 \times 3 = 12$ **[A1]**

[Back to Question] *[Total Marks: 10]*

28 $\sqrt{75} = 5\sqrt{3}$ **[B1]**

$\sqrt{27} = 3\sqrt{3}$ or $\sqrt{12} = 2\sqrt{3}$ **[B1]**

$6\sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 3]*

29 **(a)** $12 - 3\sqrt{2} + 4\sqrt{2} - 2$ **[M1]**

$10 + \sqrt{2}$ **[A1]**

(b) Multiply numerator and denominator by $(5 + \sqrt{3})$ **[M1]**

Denominator evaluates to $25 - 3 = 22$ **[M1]**

$5 + \sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

30 **(a)** $\frac{1}{2} \times (2 + \sqrt{3}) \times h = 5 + 3\sqrt{3}$ **[M1]**

$h = \frac{10 + 6\sqrt{3}}{2 + \sqrt{3}}$ **[M1]**

Multiply numerator and denominator by $(2 - \sqrt{3})$ **[M1]**

$h = \frac{20 - 10\sqrt{3} + 12\sqrt{3} - 18}{4 - 3} = 2 + 2\sqrt{3}$ **[A1]**

(b) $y^2 = (2 + \sqrt{3})^2 + (2 + 2\sqrt{3})^2$ **[M1]**

$y^2 = (4 + 4\sqrt{3} + 3) + (4 + 8\sqrt{3} + 12)$ **[M1]**

$23 + 12\sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 7]*